

Article

The Development of Synchronous Motor Using Non-Rare Earth Permanent Magnet to Improve Environmental and Economic Sustainability

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Abstract. In electric vehicles, permanent magnet synchronous motors (PMSMs) are widely used due to their high torque and efficiency. They commonly use rare earth materials, neodymium, as permanent magnets. These come with high costs and environmental impacts from their production processes. This research aims to design PMSM motors by replacing neodymium magnets with non-rare earth materials, manganese-bismuth (MnBi) and ferrite. Firstly, the requirements torque, and rotational speed of the motor were calculated under extreme conditions for carry loads of 2.5 tons. Next, the ANSYS MotorCAD software was used to adjust the motor including replacing the neodymium with MnBi and ferrite, followed by changing the magnet embedding pattern, increasing the magnets size, and increasing the overall motor size. Finally, two models were selected to compare with the neodymium motor. The results showed that the motors, using MnBi magnets, could provide torque more than ferrite. They achieved the requirements by using delta shaped as magnets embedded pattern and increasing the overall motor size, while providing lower cost and carbon emissions than the original. In conclusion, the PMSM motors, using MnBi magnets, can be sustainable to use in electric vehicles. Its potential can reduce reliance on rare-earth minerals while still meeting practical performance requirements.

Keywords: electric vehicles, permanent magnet synchronous motors, rare earth materials, non-rare earth materials, carbon emissions, sustainable

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1. Introduction

The global environment is continuously deteriorating, leading both producers and consumers in the industrial sector to increasingly prioritize pollution reduction and focus on sustainable development. This is especially true in the automotive industry, which is undergoing a major transformation. Fossil fuel for vehicles is a trend to be expensive and reduce massively; moreover, it causes air pollution due to combustion. Electric vehicles are one of the solutions to reduce fuel consumption, pollution, and fuel cost. From the car registration recorded by Thailand Automotive Institute during 2018 to 2023, the proportion of electric vehicle registrations increased by 9%, with a strong upward trend expected in the future [1]. This indicates a growing consumer preference for electric vehicles due to their ability to reduce air pollution and offer lower fuel costs compared to gasoline cars.

In Thailand, pickup trucks have consistently been the most popular vehicle type and play a crucial role in driving the economy, particularly in logistics and transportation. According to statistics on vehicle registrations by usage type from the Thailand Automotive Institute [1], small pickup trucks account for over 50% of all vehicle types. The pickup trucks are highly favored for their versatile applications because of their power designed for highly capable and heavy load. Nowadays, in logistics, electric pickup trucks tend to be used due to reducing cost that electricity is cheaper than fuel. Their power is close to a mini truck. For example, CP-Foton TM-blue45, a mini truck, has an electric motor which generates 80 kW (108 hp) and maximum torque of 220 N-m [2].

The electric motor is the core of an electric vehicle (EV). Currently, popular motor types include brushed direct current motor (Brushed DC), brushless direct current motor (BLDC), permanent magnet synchronous motor (PMSM), and induction motor (IM) [3, 4]. Among these, the PMSM is the most widely favored due to its high torque and efficiency [5]. However, PMSMs commonly use neodymium (NdFeB) permanent magnets, which are rare earth materials. These come with high costs and significant environmental impacts from their production processes [6 -12].

As stated in "The Consequences of China's New Rare Earths Export Restrictions" [13], NdFeB is currently the most powerful permanent magnet, produced from rare-earth elements (REEs), a critical component in electric vehicle systems. Furthermore, China is the world's largest producer and exporter of REEs, accounting for over 90%. In the future, there may be affected by economic volatility and trade wars between nations, leading to supply chain uncertainty. ~~If China restrict REE exports, it could severely damage the electric vehicle motor manufacturing industry.~~

In contrast, non-rare earth permanent magnets like manganese-bismuth (MnBi) and ferrite offer lower costs, require less energy to produce, and have a reduced environmental footprint. While their residual magnetic flux density is lower than NdFeB, they still hold sufficient

potential for EV applications. Notably, MnBi has a residual magnetic field intensity that is approximately 1.72 times weaker than that of neodymium, as highlighted in the research "Trade study for rare-earth-free interior permanent magnet synchronous machine using MnBi permanent" [6]. This research explored designing and simulating IPMSMs, replacing NdFeB with MnBi to lessen reliance on rare earth materials. The study compared various scenarios: Case A used NdFeB as a reference, while Cases B and C utilized MnBi magnets, with adjustments to their size, shape, and quantity to achieve performance comparable to NdFeB. The finite element analysis (FEA) was employed to analyze factors like average torque and overall motor efficiency. However, no existing research has focused on developing PMSM motors with MnBi and ferrite for practical, real-world applications.

This leads to our current endeavor: developing a practical PMSM motor by applying knowledge from relevant research, specifically targeting its use in multi-purpose pickup trucks. A survey of driver usage data and market-available vehicles indicates that these trucks typically carry loads with a total weight of no more than 2.5 tons. Given their popularity and diverse daily applications, our focus is on developing an interior permanent magnet synchronous motor (IPMSM). This type of PMSM is well-suited for pickup trucks because these vehicles demand high torque for their operational needs, and IPMSMs can deliver high efficiency and torque even at low speeds [14], which is ideal for this vehicle type. The calculation showed that to cover all load conditions, the motor must provide at least 200 N-m of torque and achieve a speed of at least 9,000 rpm to be sufficient for these trucks.

This research aims to investigate the impact on efficiency and sustainability by replacing NdFeB magnets in PMSMs with MnBi and ferrite. The analysis was performed using specialized Finite Element Analysis software, ANSYS MotorCAD, which can simulate and analyze motor torque, power loss, and various performance aspects. Different motor geometries were simulated and adjusted to observe torque trends. The IPMSM motors were also scaled using MnBi and ferrite instead of NdFeB to assess whether non-rare earth magnets can generate sufficient torque for a 2.5-ton pickup truck.

Beyond performance, this research also assessed environmental sustainability, including the impacts of mineral extraction, processing, and carbon emissions. Furthermore, we examined economic stability and supply chain security by reducing reliance on rare earth minerals from primary producing countries like China [13]. Regarding costs, MnBi is approximately 11 USD/kg and ferrite is 10 USD/kg, while NdFeB is significantly more expensive at 25 times higher [6]. Therefore, this research aims to develop non-rare earth permanent magnet motors that offer comparable or near-comparable performance to conventional types, while also being more sustainable and environmentally friendly.

2. Method

The study began by establishing the problem statement: developing a PMSM motor by replacing the permanent magnet material in the rotor from NdFeB (rare-earth magnet) to MnBi and ferrite (non-rare-earth magnet). The experiment determined if, after changing the magnet type, the motor can be used in a multi-purpose pickup truck with a total payload not exceeding 2.5 tons.

Next, a benchmark motor from the market was identified. For this research, motor specifications, specifically those of the 2012 Nissan Leaf motor [15], were referred to as creating a benchmark motor, or prototype, for this report. The prototype motor was defined to use NdFeB magnet material and has a minimum torque of 220 N-m. Therefore, the prototype motor had adjusted its torque down to approximately 220 N-m, before it was served as the baseline motor for comparing experimental results in this research.

2.1. Torque and angular velocity calculation for utilizing

This section details the calculations for the required torque and angular velocity the motor must produce for an electric pickup truck of the same size as our baseline, which is a multi-purpose pickup truck with a gross laden weight of 2.5 tons. The design criteria for this electric multi-purpose pickup truck were calculated in two cases to determine the required power and torque.

2.1.1. The required angular velocity for the motor

To calculate required angular velocity for the motor, parameters were defined including wheel radius ($r = 0.3$ m), final drive ratio ($n = 10.5$), the maximum speed ($v_{\max}$) referred to as the maximum legal speed for this type of vehicle (not exceeding 90 kilometers per hour [16]). From Eq. (1), the angular velocity of the wheel (ω_{wheel}) had been calculated, then it was substituted to Eq. (2) to get angular velocity for the motor (ω_{motor})

$$\omega_{\text{wheel}} = \frac{v_{\max}}{r} \quad (1)$$

$$\omega_{\text{motor}} = \omega_{\text{wheel}} \times n \quad (2)$$

2.1.2. The required torque for the motor

To determine the torque requirements for designing an electric vehicle motor, the forces the vehicle needs to generate to overcome various road loads were considered. The extreme case that encompasses all loading conditions is driving uphill, as illustrated in Fig. 1. The forces required for vehicle propulsion can be determined from road loads [17-20] including drag force (R_d), rolling resistance between wheels and road (R_r), gradient resistance from a slope (R_g), and acceleration resistance (R_a), defined in Eq. (3)-(6) respectively. Finally, these variables are summed as total required force, or wheel force (F_w), in Eq. (7).

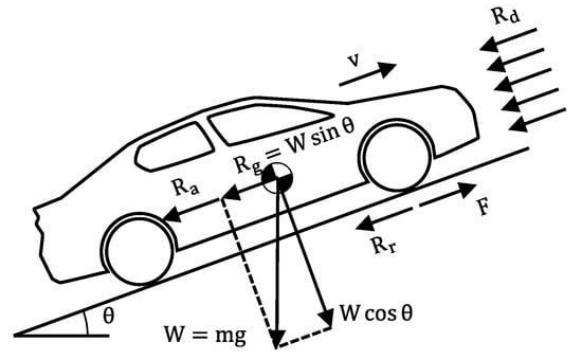


Fig. 1. Free body diagram of car in motion under the maximum load conditions [20].

$$R_d = 0.5 C_d \cdot A \cdot \rho \cdot v^2 \quad (3)$$

$$R_r = C_{rr} \cdot m \cdot g \cdot \cos \theta \quad (4)$$

$$R_g = m \cdot g \cdot \sin \theta \quad (5)$$

$$R_a = m \cdot a \quad (6)$$

$$F_w = R_d + R_r + R_g + R_a \quad (7)$$

When

C_d	drag coefficient (0.4)
A	the frontal area of a car (2.34 m ²)
ρ	the density of the air (1.225 kg/m ³)
v	the relative velocity of the object (30 km/hr = 8.33 m/s)
C_{rr}	the coefficient of rolling resistance (0.03)
m	gross vehicle mass (2,500 kg)
g	gravitational acceleration (9.81 m/s)
θ	slope angle or gradeability (12 degrees)
a	car acceleration

Gradeability is set at 12%, accounted for in the design as an allowance although the law stipulates that the slope of a hill should not exceed 6.84 degrees or 12% [21]. This is because the incline significantly impacts the resistive forces. Besides, acceleration resistance is zero since we consider a constant speed of 30 kilometers per hour. When all forces are combined, the wheel force was used to calculate the torque at the wheel (T_w) by multiplying the wheel radius, 0.3 m. Then, the wheel torque was divided by the final drive ratio of 10.5 to get the required motor torque (T_m).

2.2. ANSYS MotorCAD 2024 verification

This research utilizes ANSYS MotorCAD 2024. For accurate program operation, experimental procedures, and data acquisition, experiments were performed in accordance with "Analysis and Optimization of IPMSM for EV Applications Using Motor-CAD" by Ayhan, D. [22]. This report details the simulation of an interior permanent magnet synchronous motor (IPMSM) for electric vehicles, the 2012 Nissan Leaf. This includes specifying the rotor and stator dimensions, magnet configuration, winding details, wire gauge, and other parameters. The ANSYS MotorCAD software was then used to run simulations with these input parameters, and the resulting data was compared to ascertain their proximity.

2.3. Designing a baseline motor

In this research, Nissan Leaf motor was chosen as the baseline motor to compare results. However, due to limitations in the specifications and size of motors available for development, this motor was reduced the torque to be closer to benchmark vehicle, the CP FOTON™ iBlue45 multi-purpose pickup truck, which has a maximum torque of approximately 220 N-m. To adjust the motor, referred to as Eq. (8), the motor length directly affects its volume and, consequently, its maximum torque output. As illustrated in Fig. 2 and 3, the torque reduction was done by decreasing the length of the motor. Originally, the motor stator length was 160 mm, and the rotor and magnet length were 150 mm. After iterative simulations to reduce the length of the stator, rotor, and magnet sections, the new stator length was 130 mm, and the new rotor and magnet length was 120 mm.

$$T_e = \frac{\pi}{4} D_r^2 l_e B_g K_s \eta_r p_f \quad (8)$$

When

T_e	electromagnetic torque (N-m)
B_g	airgap flux density (Tesla)
K_s	peak linear current density (A/m)
D_r	rotor diameter (m)
l_e	effective motor length (m)
η_r	rotor efficiency
p_f	power factor

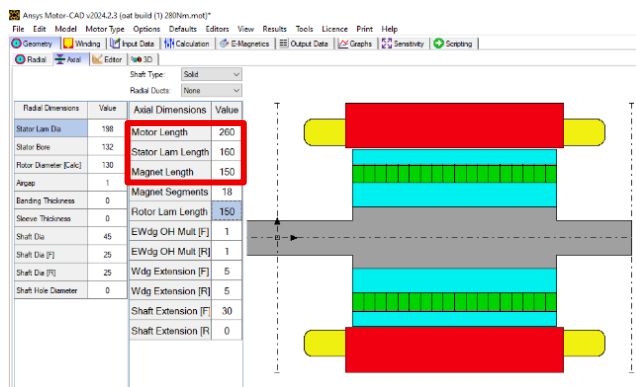


Fig. 2. The original motor geometry.

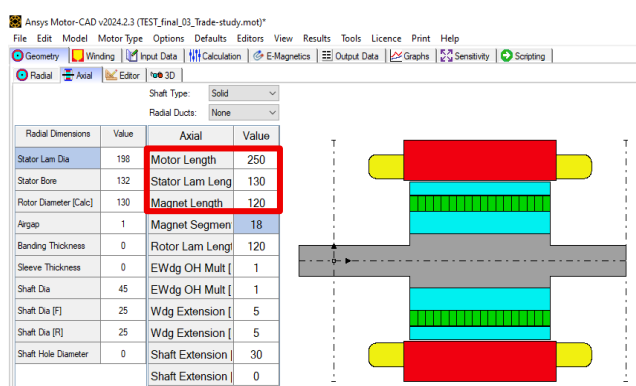


Fig. 3. The adjusted motor geometry.

2.4. Modifying magnet materials in the PMSM model (MnBi and ferrite)

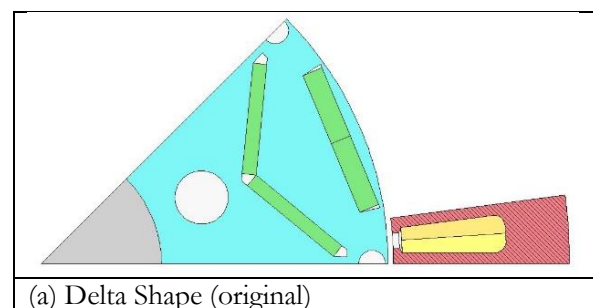
Although the obtained motor has torque output which is comparable to the benchmark vehicle, it was adjusted by using rare-earth permanent magnets, neodymium (NdFeB) magnets installed within the rotor. This research aims to investigate the effects of replacing rare-earth neodymium (NdFeB) magnets with two types of non-rare-earth magnets, manganese bismuth (MnBi) and ferrite (Y32). During this experiment, other motor parameters and component dimensions remained unchanged. A newly designed motor had been simulated using ANSYS MotorCAD, and compared the results to see how the torque changes after modifying the permanent magnet material.

After replacing the rare-earth permanent magnet material (NdFeB) with non-rare-earth, manganese bismuth (MnBi) and ferrite (Y32), it was observed that MnBi yielded significantly lower torque than NdFeB, but it still produced a higher torque level compared to ferrite (Y32).

Consequently, MnBi is deemed a more suitable material for further development. This is due to its superior magnetic properties, such as higher remanence (Br) and coercivity (Hc) values when compared to ferrite. As a result, the motor utilizing MnBi has the potential to generate higher torque.

2.5. Modifying the permanent magnet embedding pattern in the rotor

The permanent magnet embedding pattern within the rotor directly influences magnetic field density, and torque output. This research additionally designed the three motor models with different embedding patterns in the rotor, as detailed in Fig. 4. These models were applied only to MnBi magnets, additionally, the weight of the magnets was controlled and maintained within a tolerance of 0.010 kg. Subsequently, the simulation was conducted to observe the trend of torque changes for each different permanent magnet embedding pattern in the rotor.



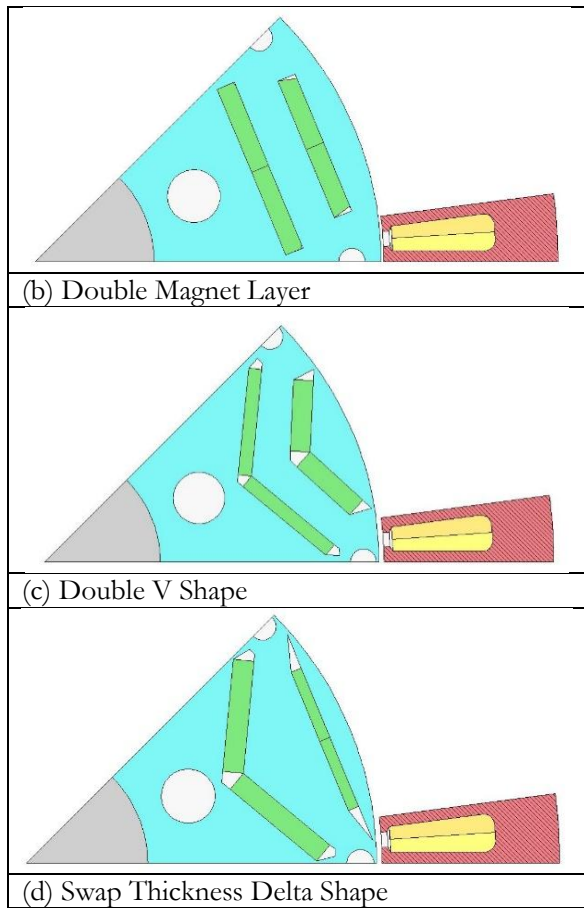


Fig. 4. The embedding pattern of permanent magnet in the rotor.

The delta shape (a), original from Nissan leaf, the inner magnets are embedded in a V-shape, while the outer magnets are embedded linearly along the rotor's axis. The outer magnets are thicker than the inner magnets.

The double magnet layer (b) involves embedding permanent magnets in two layers, both following a straight, linear arrangement along the rotor's axis. Both the inner and outer magnet layers consist of MnBi permanent magnets, each with a thickness of 3.5 mm.

In double V shape (c), the magnets are embedded in V-shape in both inner and outer layer. The swap thickness delta shape (d) is similar to the delta shape, but the inner magnets are thicker than the outer magnets.

After designing the embedding pattern of permanent magnet in the rotor, each model was checked its weight close to the original model within a tolerance of 0.010 kg. Then, the simulation in ANSYS MotorCAD was set up with the following electrical load parameters including voltage of 375 V, peak current of 480 A, and phase advance of 51.83 degree.

2.6. Motor performance improvement approaches

To enhance motor's torque output to meet the minimum torque requirement of 200 N-m, this research explored and identified additional methods in two ways.

2.6.1. Increasing the overall motor size

The motor torque is directly proportional to the rotor's volume which is directly related to both its diameter and length [19]. This experiment involved scaling up the motor size by percentages of 100% (original value), 110%, 120%, 130%, and 140%. Simultaneously, the rotor length increased, starting from 120 mm (original value) to 300 mm. The scaling-up method involved multiplying the dimensions of the rotor and stator diameters, as well as the dimensions of the magnetic components within the rotor, by the percentage increase, as presented in Table 1.

Table 1 Simulation Parameters for Stator Diameter and Rotor Length.

Parameters	Values (mm)
Stator diameter	198, 217.8, 237.6, 257.4, 277.2
Rotor length	130, 145, 160, 175, 190, 205, 220, 235, 250, 265, 280, 295, 300

The values for increasing the motor's length and expanding its diameter were determined primarily by considering the actual installation space within the vehicle. The benchmark vehicle used for comparison is the CP-Foton TM-iBlue multi-purpose pickup truck. As shown in Fig. 5, measurements indicated that there is still available space for increasing both the length and diameter of the motor, but the maximum length (L) and diameter (D) must not exceed 300 and 315 mm respectively. Additionally, from Benchmarking State-of-the-Art Technologies [15], it was noted that motor housing typically has a thickness of 20 mm.

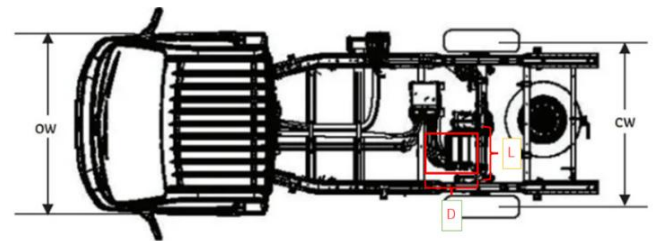


Fig. 5. Size and position of motor housing in the CP-Foton TM-iBlue.

After performing simulations for each case, 65 scenarios (5 scaling percentages and 13 rotor lengths), experimental data was analyzed by plotting it on a 3D graph under the conditions of the maximum allowable motor length and maximum allowable motor radius that had been previously established. This plot helped determine the optimal motor size to achieve the target torque.

2.6.2. Increasing the size of the permanent magnets

According to reference [22], increasing the magnet parameters (width, angle, and thickness), while decreasing the air gap length, increase the torque output. Specifically:

- Increasing magnet dimensions lead to higher torque but also affect other parameters like power loss, torque ripple, and magnet weight.
- Decreasing the air gap length reduces the space between the rotor and the stator, allowing magnetic flux to flow more easily and thus increasing torque.

This section adjusted the following permanent magnet parameters within the rotor, detailed in Fig. 6 including inner (L1) and outer (L2) magnet bar width, L1 pole V-angle, L1–L2 magnet thickness, and the air gap length. The embedded pattern, used in the experiment, was chosen from the pattern which provided the highest torque, done in section 2.5.

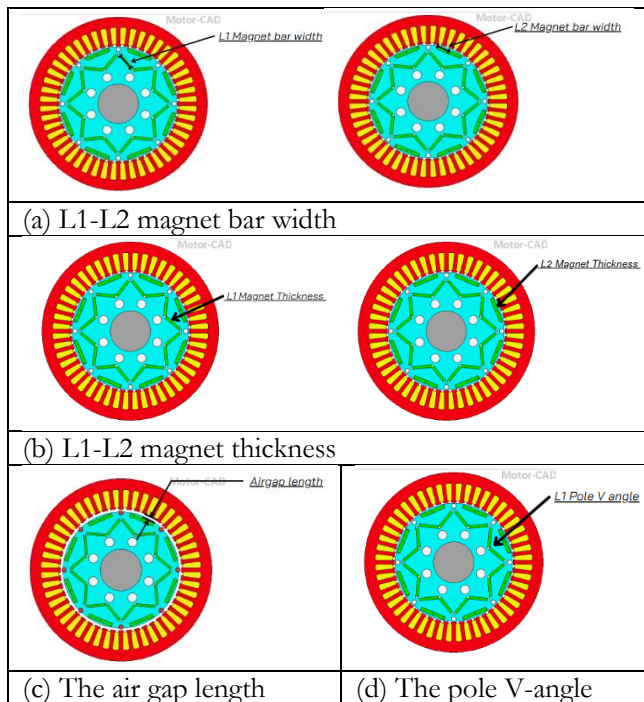


Fig. 6. The location adjusted permanent magnet parameters.

As a result, the values for parameters including the magnet bar width, L1 pole V angle, and the magnet thickness were each increased by two distinct values. Conversely, the air gap length parameter decreased by two distinct values, as detailed in Table 2. However, all parameter adjustments must be made with caution and within the physical limits of the design. The changes must not conflict with the size of other components, such as other magnet layers or the shaft.

Table 2 Simulation parameters for magnetic geometry adjustment

Parameter	Unit	Original Value	Value For Simulation
L1 Magnet Thickness	mm.	2.6	2.6, 2.8, 3.0
L2 Magnet Thickness	mm.	3.862	3.862, 4, 4.2
L1 magnet bar width	mm.	20.88	20.88, 21, 21.2
L2 magnet bar width	mm.	13.9	13.9, 14.1, 14.3
L1 Pole V angle	Degree.	124	124, 128, 132
Airgap length	mm.	1	0.9, 0.95, 1.0

Next, the designed parameters were entered into the ANSYS MotorCAD, Sensitivity function, as shown in Fig. 7. This experiment simulated a total of 729 cases (3 values per variable for six variables) with electrical load. For all simulations, the output measurements were taken at 2750 RPM

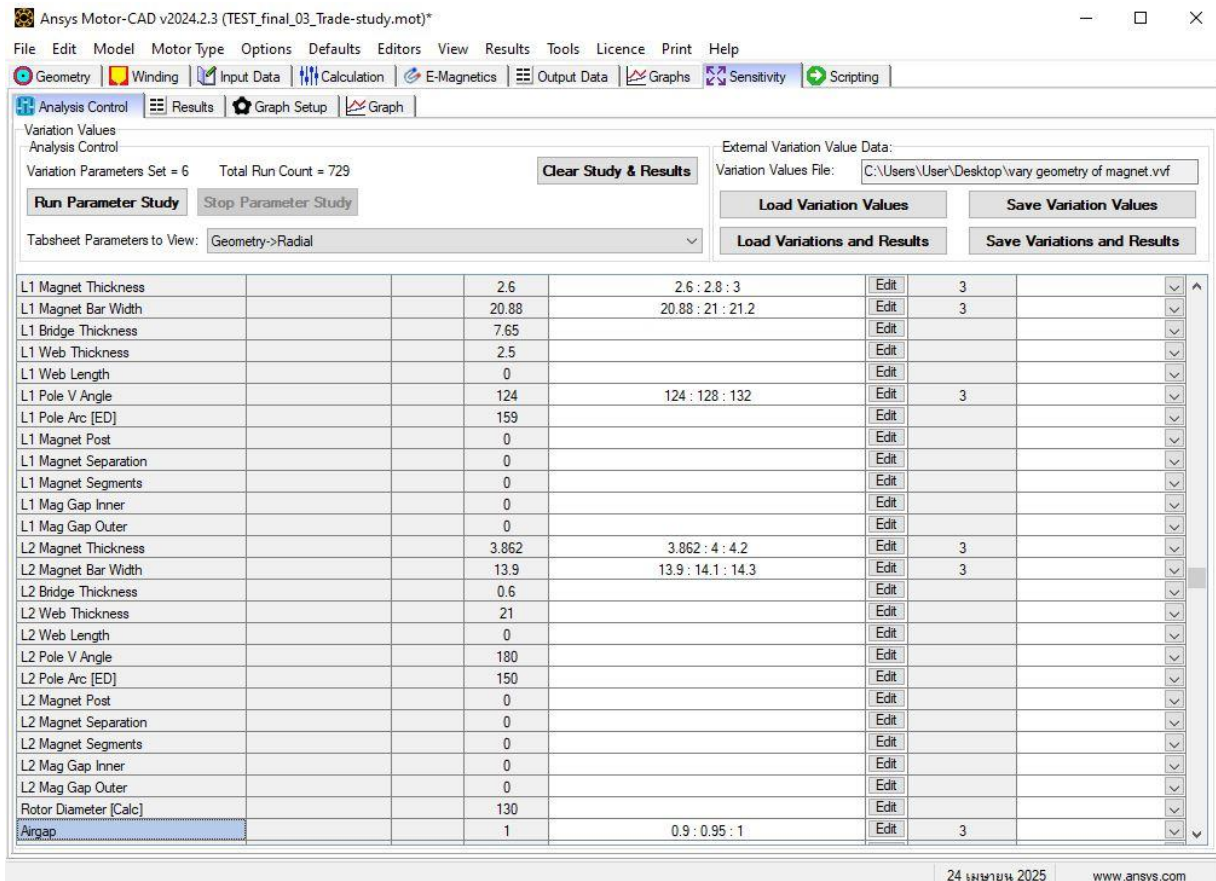


Fig. 7. Setup window for Sensitivity function in ANSYS MotorCAD

2.7. Motor design considering performance with other criteria

Based on the study by Brody et al. (2022) [6], increasing the volume of a PMSM could enable it to meet the required torque criteria. In this section, appropriate PMSM motor sizes within defined dimensional limits were selected from several PMSM motor sizes that met the criteria. Hence, three different PMSM motor models were selected and compared to fit for multi-purpose pickup truck.

- Model 1 uses a rare-earth element, NdFeB, as magnets.
- Model 2 was chosen from those providing the target torque of 200 N-m, and had the smallest diameter. The smaller diameter increases the ground clearance, reducing the risk of the motor hitting the road or other environmental obstacles.
- Model 3 provided the target torque of 200 N-m, and had the largest diameter, as a larger diameter simplifies the installation of additional cooling systems.

These selected motors had been scored across five categories, as detailed in Table 3, before this was a weighted scoring comparison, with scores ranging from 1 to 5, as shown in Table 4. The criteria for comparison between models include:

- Motor efficiency

- Torque ripple: The fluctuation in torque output.
- Total weight: Directly obtained from experimental data.
- Cost: Calculated based on the weight of the rotor, stator, copper wire, magnets, and shaft, multiplied by the cost of each component, as shown in Table 5.
- Carbon emissions: Calculated based on the weight of the magnets used in each motor, as magnet production is a primary source of emissions. Due to limited data, carbon emissions were specifically compared only during the mining phase to demonstrate the general trend of carbon emission quantity, with values referenced from Table 6.

Table 3 Motor properties scoring ranges

Items	Score	Indicator
Efficiency	5	$90\% \leq \text{Efficiency} \leq 100\%$
	4	$80\% \leq \text{Efficiency} \leq 90\%$
	3	$70\% \leq \text{Efficiency} \leq 80\%$
	2	$60\% \leq \text{Efficiency} \leq 70\%$
	1	$50\% \leq \text{Efficiency} \leq 60\%$
Torque ripple	5	Torque ripple $\leq 5\%$
	4	$5\% < \text{Torque ripple} \leq 10\%$
	3	$10\% < \text{Torque ripple} \leq 15\%$

	2	$15\% < \text{Torque ripple} \leq 20\%$
	1	Torque ripple $> 20\%$
Weight	5	Weight ≤ 20 kg
	4	$20 \text{ kg} < \text{Weight} \leq 25$ kg
	3	$25 \text{ kg} < \text{Weight} \leq 30$ kg
	2	$30 \text{ kg} < \text{Weight} \leq 35$ kg
	1	Weight > 35 kg
Cost	5	Cost ≤ 100 USD
	4	$100 < \text{Cost} \leq 150$ USD
	3	$150 < \text{Cost} \leq 200$ USD
	2	$200 < \text{Cost} \leq 250$ USD
	1	Cost > 250 USD
Carbon emission	5	Carbon ≤ 2 gCO ₂ /kg
	4	$2 < \text{Carbon} \leq 4$ gCO ₂ /kg
	3	$4 < \text{Carbon} \leq 6$ gCO ₂ /kg
	2	$6 < \text{Carbon} \leq 8$ gCO ₂ /kg
	1	Carbon > 8 gCO ₂ /kg

Table 4 Weighting factors for each criterion

Criteria	Weight
Efficiency	0.1
Torque ripple	0.2
Weight	0.2
Cost	0.2
Carbon emission	0.3
Total score	1.0

Table 5 Material cost for components in a motor

Component	Material	Price (USD/kg)
Stator	M250-35A	1.50
Rotor	M250-35A	1.50
Wire	Copper	11.00
Magnet	NdFeB	27.42
	MnBi	26.00
	Ferrite	3.00
Shaft	Aluminium-alloy-cast	3.10

Table 6 Global Warming Potential (GWP) from mining and production [23-25]

Magnet	Mineral	GWP from production (kg CO ₂ -eq)
NdFeB	NdFeB	27
MnBi	Mn	19.825
	Bi	
Fe	Fe	9.7

3. Results and Discussion

3.1. Calculated minimum torque and rotational velocity for the electric pickup truck

Based on the calculations of required motor speed and torque in section 2.1, the designed motor must generate a minimum torque of 191 N-m to be sufficient to handle the extreme case load. Additionally, the motor needs an angular velocity of 8356 RPM to support driving at a maximum speed of 90 km/hr. Thus, this study identified the minimum torque and maximum angular velocity provided to the motor as 200 N-m and 9000 RPM respectively; these values are close to the electric multi-purpose pickup trucks in commercially available vehicles.

3.2. Validation of ANSYS MotorCAD 2024

Obtained the following output parameters from the simulation including shaft torque, output power, and efficiency, these values were read at a motor speed of 2750 RPM and then compared with the results from the reference document, Analysis and Optimization of IPMSM for EV Applications Using Motor-CAD [22]. After the simulation had been completed, the results showed that the maximum torque was 280.3 N-m. The output power was 80.7 kW, and the efficiency was 94.73%. When comparing the experimental data with the reference source in Table 7, it was found that the simulation results are very closely aligned, with errors less than 1%. This is considered acceptable and demonstrates that the software usage is accurate and reliable.

Table 7 Comparison of output parameters from the simulation with the reference [22]

Output values	Reference values [22]	Values from the Simulation	Error (%)
Torque (N-m)	279	280.3	0.466
Power (kW)	80	80.7	0.875
Efficiency (%)	94.69	94.73	0.04

3.3. Adjustment of baseline motor

Figure 8 shows the relation between torque and rotational speed when the torque was reduced by decreasing motor length using ANSYS MotorCAD. Originally, with a rotor length of 150 mm, the motor produced 280 N-m of torque at an operating speed of 2750 RPM. When the rotor length was reduced to 120 mm, a 20% decrease in length, the motor was able to produce 224 N-m of torque. This 20% reduction in torque directly corresponds to the 20% reduction in rotor length, consistent with Eq. (8).

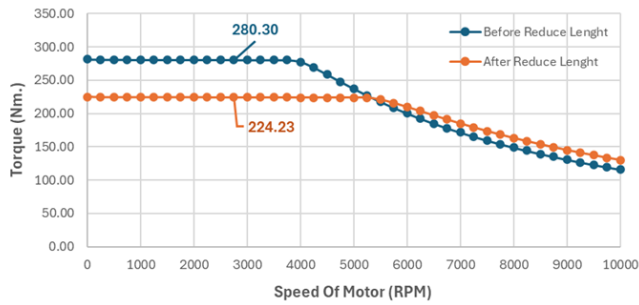


Fig. 8. The relationship between torque and rotational speed before and after length Reduction.

3.4. Results of permanent magnet material modification

As shown in Fig. 9, after changing the magnet material within the rotor, the baseline motor, which utilizes a rare-earth magnet (NdFeB), produced a torque of 224 N-m at 2750 RPM. This is significantly higher than motors using non-rare earth magnets; MnBi yielded 166 N-m, and ferrite (Y32) yielded 131 N-m.

The result shows that the motor's torque output decreased by approximately 50–90 N-m. This was affected by the remanence values, which indicate the strength of the permanent magnet. NdFeB possessed the highest magnetic strength, followed by MnBi, and then ferrite. Consequently, the torque generated by the motors with non-rare-earth magnets was insufficient to meet the minimum torque requirement of 200 N-m for a multi-purpose pickup truck with a gross weight not exceeding 2.5 tons. Therefore, this research primarily used MnBi as non-rare-earth permanent magnets for further development due to providing torque higher than ferrite.

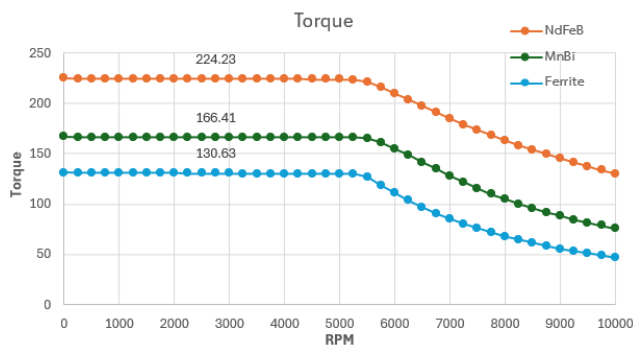


Fig. 9. The PMSM motor torque for different permanent magnet materials.

3.5. Experimenting with permanent magnet embedding patterns in the motor rotor

Based on the torque values for each embedding pattern shown in Fig. 10, it identifies that the delta shape (the original design) yielded the highest torque at 166.41 N-m. Furthermore, Figure 11 indicates that the delta shape also had the highest efficiency at 92.46%, making it the most efficient among the four magnet embedding

patterns followed by the double V shape, the swap thickness delta shape, and the double magnet layer orderly. It can be said that the embedded pattern affected flux linkage which has directly changed to torque. However, the torque from the best embedded pattern was less than the minimum torque requirement, 200 N-m.

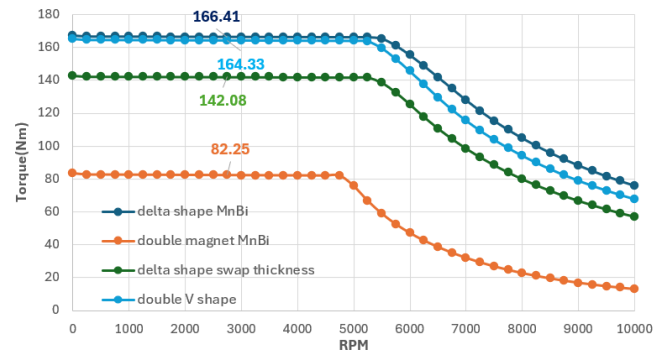


Fig. 10. The motor torque for different permanent magnet embedding patterns.

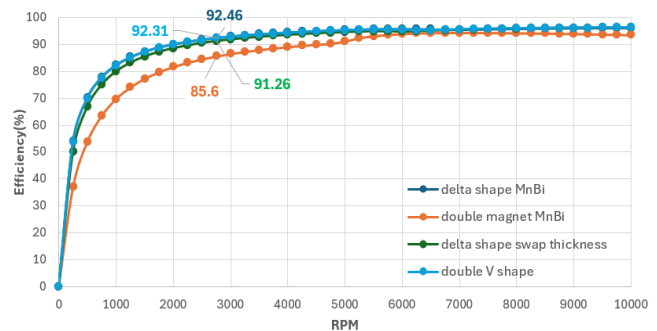


Fig. 11. Motor efficiency for different permanent magnet embedding patterns.

3.6. Performance improvement

From section 2.6, to improve motor torque, there were 2 options considered. Varying the parameters of the permanent magnets was prioritized as the first to test, while increasing the overall motor size was followed if the first option cannot give the minimum required torque.

3.6.1. Increasing the size of the permanent magnets

After varying parameters including L1 magnet bar width, L2 magnet bar width, L1 pole V angle, L1 magnet thickness, L2 magnet thickness, and Air Gap Length, the total of 729 simulations were analyzed, as outlined in Section 2.7.

The results indicate that adjusting the geometry of the MnBi permanent magnets in the rotor directly impacted the total amount of permanent magnet material in the motor. Increasing the size of the permanent magnets led to a higher magnet weight, and increased torque.

However, the graph in Fig. 12, the X-axis represents the weight of the permanent magnets, and the Y-axis shows the motor's torque output, notice that the highest

torque achieved was 173.1 N-m. This occurred when the magnet weight in the rotor was at its maximum, 1.852 kg. Therefore, it shows that adjusting the geometry of the permanent magnets in the rotor changed the torque output as an increasing trend, but it is still insufficient for use in a multi-purpose pickup truck with a gross weight not exceeding 2.5 tons. Since this option did not achieve its goal, the next option was performed.

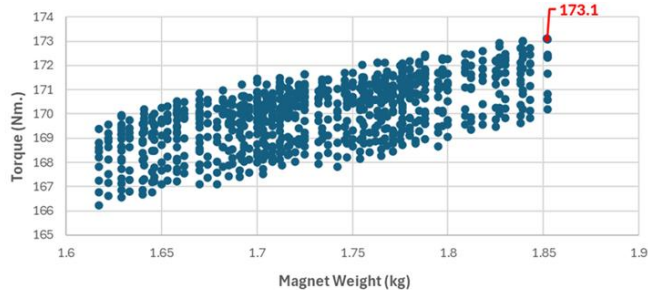


Fig. 12. The relationship between torque and permanent magnet weight in the rotor for all 729 cases.

3.6.2. Increasing the overall motor size

This experiment was on increasing the torque values obtained by varying motor's diameter from 198 to 277 mm, and the motor length from 130 to 300 mm. As detailed in Fig. 13, the torque consistently increased across all increased lengths and diameter. The torque started at 166 N-m and peaked at 625 N-m. This clearly indicates that an increase in overall dimensions significantly boosts torque output. Furthermore, the result shows a clear trend of exceeding 200 N-m of torque without needing to push beyond the defined dimensional limits.

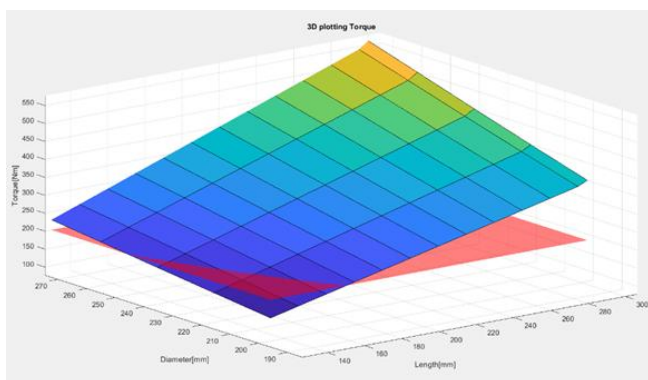


Fig. 13. 3D-plot shows required torque (the red plane), and the trend of torque corresponding with increasing diameter and length (the contour plane).

Table 8 Calculated parameters for evaluation

	System Efficiency (%)	Torque Ripple (MsVw)	Total Motor Weight (kg)	Total Prices. (USD)	Carbon Emission (kg CO ₂)
Model 1	94.23	16.39	27.04	225.04	7.15
Model 2	92.98	11.16	31.75	127.28	3.72
Model 3	92.99	11.54	31.80	123.65	3.75

Analyzing the motor's torque at the target of 200 N-m, which is the acceptable minimum torque, as monitored in Fig. 14, plotting the intersection points of torque output with motor overall dimensions illustrates the relationship between motor diameter and length. The graph clearly divides into two regions: The area above the line represents the range where the motor can produce more than the minimum required 200 N-m of torque. The area below the line indicates where the motor's torque output is less than the requirement. On the line, it demonstrates the motor sizes which deliver precisely 200 N-m of torque. These dimensions can be used as a criterion in designing or selecting motors that precisely meet the desired specifications.

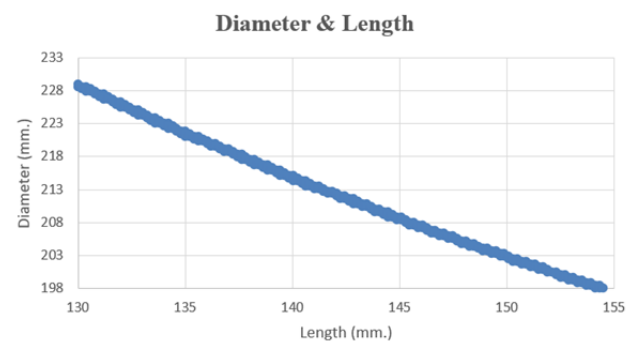


Fig. 14. The relationship between motor diameter and length for all motors capable of producing 200 N-m of torque.

3.7. Comparison of designed motor in other criteria

The goal of this comparison is to identify a motor suitable for a multi-purpose electric pickup truck. The evaluation was conducted according to Table 3 in section 2.7. The first model was the baseline motor using NeFeB. The other models were chosen from considering the graph in Fig. 14. One was the motor which has the smallest diameter, another was the largest diameter.

After the three models were selected, the simulation was run again, and the result was shown in Table 8. Finally, each model was scored as detailed in Table 9.

Table 9 Comparing the models with weighting score

Items	Weight	model#1	model#2	model#3
Efficiency	0.1	5	5	5
Torque ripple	0.2	4	5	5
Weight	0.2	5	3	3
Cost	0.2	4	5	5
Carbon emission	0.3	3	5	4
Total score		4	4.6	4.3

When comparing the data in Table 9, using key indicators including efficiency, torque ripple, weight, cost, and carbon emissions, each motor possesses distinct characteristics.

Model 1, which uses NdFeB magnets, was scored highly in efficiency and weight, achieving full marks in both categories. However, this model performed poorly in terms of cost and carbon emissions, resulting in an overall score of 4.

Model 2, designed with a smaller diameter to reduce the risk of striking the road surface, was scored very highly in torque ripple, cost, and carbon emissions. Its high score in carbon emissions highlights its potential for reduced environmental impact. Due to this balanced set of factors, model 2 achieved the highest overall score of 4.6, indicating that it is the most suitable motor for use in the multi-purpose electric pickup truck.

Although, model 3, featuring the largest diameter, was scored well in cost and carbon emissions, similar to model 2, it had a notable drawback in weight, ranking the lowest, and its efficiency was not particularly high, leading to an overall score of 4.3.

4. Conclusion

This research explores the replacement of rare-earth NdFeB permanent magnets with more sustainable alternatives, MnBi and ferrite, targeting a desired motor torque of 200 N-m.

Using ANSYS MotorCAD for design and simulation, it was found that neodymium (NdFeB) magnets yield the highest torque at 224 N-m. Motors using MnBi showed the closest performance in terms of torque and applicability to NdFeB, achieving 166 N-m when employing the delta shape magnet embedding pattern, which provided the highest torque among tested configurations. Ferrite, however, had significant torque limitations, resulting in the lowest torque at 131 N-m. This confirmed that among non-rare-earth magnets, MnBi offers the highest torque output.

Consequently, this study focused on MnBi motors and attempted to increase their torque through magnets geometry adjustments. This only resulted in an increase of 6.69 N-m, which was insufficient for the application. Then, increasing the torque by enlarging its diameter or length was performed. This method successfully demonstrated

that increasing the motor's volume can achieve the desired 200 N-m torque.

Furthermore, an environmental analysis using Global Warming Potential (GWP), which measures carbon emissions, alongside other indicators: efficiency, torque ripple, weight, and cost, were considered with the three selected models. The weighted scoring results indicated that models 2 and 3 are viable replacements for model 1. When considering the balance of various parameters, particularly cost and sustainability, models 2 and 3 scored higher than model 1, and model 2 (small diameter) achieved the highest overall score. It seems that the motor using MnBi permanent magnets can be sustainable to use in electric multi-purpose pickup trucks. This is possible that its potential can reduce reliance on rare-earth minerals while still meeting practical performance demands.

For future research directions, firstly, the carbon emission data from the mining process, which represents only a portion of magnet production, remains incomplete, particularly for manganese (Mn) and bismuth (Bi) used in MnBi magnets. It should focus on collecting real-world field data or developing more accurate Life Cycle Assessment (LCA) models to comprehensively evaluate environmental impacts. Secondly, although the remanence (Br) and coercivity (Hci) values for MnBi from theoretical studies appear viable, industrial-scale synthesis of MnBi is not yet a reality. It's possible that the magnetic properties achieved in actual production might be lower than theoretical values. Thus, the future work should include experimental production and property testing of MnBi to refine design models for closer practical application.

Lastly, this research remains at the simulation level and has not yet involved the creation of physical motor prototypes. The next step is the construction and testing of motor prototypes, especially those utilizing MnBi, to validate their performance in real-world conditions.

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